# DUDB: Deep Unfolding-Based Dual-Branch Feature Fusion Network for Pan-Sharpening Remote Sensing Images

Hailin Tao, Jinjiang Li, Zhen Hua, and Fan Zhang

***Abstract*— The proposed method aims to enhance the fusion of high-resolution multispectral (MS) images (HRMS) by extracting spatial and spectral features from panchromatic (PAN) images and MS images. However, existing pan-sharpening methods often suffer from the problem of missing spatial and spectral detail information. To better preserve these details, we introduce a dual-branch feature fusion pan-sharpening network based on deep unfolding. In this network, we utilize the algorithm unfolding iterative module (AUIF-Block) to continuously acquire detailed information from both MS and PAN images for image reconstruction. By leveraging the adaptive channel and spatial feature enhancement module (DEM-Block), the network can adjust spatial and channel features adaptively, leading to more accurate feature extraction and more complete image reconstruction. Finally, the detail-based fusion module (DBFM-Block) is employed to integrate and enrich the content of detailed information extracted from different channels, resulting in improved fusion performance. Experiments were conducted on QuickBird (QB) and WorldView-2 (WV2) datasets. Through qualitative analysis and quantitative comparisons, we demonstrate that this method outperforms existing approaches.**

***Index Terms*— Deep unfolding, multispectral (MS) images, panchromatic (PAN) images, pan-sharpening.**

## I. Introduction

IN RECENT years, with the rapid advancement of satellite sensing technology, the applications and demand for remote sensing imagery have been steadily increasing. Remote sensing images with high spatial resolution multispectral (HRMS) have become essential in various fields such as agriculture monitoring, urban planning, and geological exploration [1], [2], [3]. There is an inherent trade-off between spatial resolution and spectral resolution in remote sensing sensors [4]. Due to various physical and technical limitations of different sensors, a single sensor cannot acquire images with both high spatial and high spectral resolution [5]. Currently, remote sensing satellites provide only panchromatic (PAN) images and multispectral (MS) images. PAN images offer precise spatial information but lack spectral information, while MS images provide rich spectral information but lack spatial details [6]. To obtain HRMS images, it is necessary to fuse PAN and MS images, a process known as remote sensing image fusion, or pan-sharpening [7].

In the past few decades, traditional pan-sharpening methods have been categorized into four main approaches: component substitution (CS)-based methods [8], [9], [10], [11], [12], multiresolution analysis (MRA)-based methods [13], [14], [15], [16], variation optimization (VO)-based methods [17], [18], [19], [20], [21], and deep learning (DL)-based methods [22], [23], [24], [25].

Methods based on CS are considered classical pan-sharpening techniques. To obtain the intensity (I) component, the MS image is projected into a new feature space. To enhance spatial resolution, the PAN image is spectrally substituted, and the fusion image is obtained through an inverse transform. Representative algorithms in this category include intensity-hue-saturation (IHS) transform [26] and Gram–Schmidt adaptive (GSA) [27]. While CS-based methods efficiently preserve spatial information, they may suffer from distortion of low-frequency information in the MS image [28], [29]. Methods based on MRA primarily use multiscale techniques to extract details from the PAN image, which are then injected into the original MS image to enhance spatial resolution. Representative algorithms in this category include wavelet transforms [30] and Laplacian pyramids [31]. While MRA-based methods have significantly improved spectral distortion issues, they can be sensitive to image registration results and may produce edge artifacts [32], [33]. Methods based on VO primarily involve constructing function models by analyzing the relationships between PAN, MS, and HRMS and optimizing these models through regularization functions to achieve better results. Representative algorithms in this category include Bayesian methods [34] and sparse reconstruction methods [35]. Compared to the previous two methods, VO-based methods can yield superior fusion results, but they come with high computational costs, highly complex optimization processes, and limited optimization effectiveness.

With the development of DL, many pan-sharpening techniques based on DL have emerged and gained widespread application. Masi et al. [23] first introduced the concept of a

Manuscript received 8 September 2023; revised 24 October 2023, 13 November 2023, and 6 December 2023; accepted 6 December 2023. Date of publication 8 December 2023; date of current version 19 December 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 61772319, Grant 62002200, Grant 62202268, and Grant 62272281; in part by the Shandong Provincial Science and Technology Support Program of Youth Innovation Team in Colleges under Grant ZR2023MF026 and Grant 2019KJN042; in part by the Shandong Natural Science Foundation of China under Grant ZR2020QF012 and Grant ZR2021MF107; and in part by the Yantai Science and Technology Innovation Development Plan under Grant 2022JCYJ031. *(Corresponding author: Fan Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2023.3341076

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fully convolutional neural network for pan-sharpening, known as the pan-sharpening neural network (PNN). PNN achieved state-of-the-art (SOTA) results in its debut year. However, they noted that it lacked interpretability. For this reason, Deng et al. [36] introduced FusionNet, which combines traditional CS- and MRA-based methods with a residual network to improve interpretability. Additionally, Wu et al. [37] proposed VO + Net, which combines PNN with a variational model. These methods improved interpretability and achieved better fusion results. However, they may not fully capture spatial feature information, potentially leading to spatial distortion [30]. To enhance the quality of spatial feature information, some unsupervised pan-sharpening methods have been developed. For example, Ma et al. [38] introduced the unsupervised PanGAN method. Furthermore, some researchers have designed multibranch networks to fully exploit spectral and spatial information. Shao and Cai [39] proposed a dual-branch deep fusion network called the deep convolutional neural network (RSIFNN). This method considers the presence of redundant information between MS and PAN images. They treat the entire network as a residual unit and overlay the predicted mask onto the original MS image to obtain the fusion image. However, this method also faces challenges such as spectral distortion and contour blurring. Liu et al. [40] designed a dual-stream generalized sharpening network known as the twin-stream fusion network (TFNet). This method can separately extract multimodal features and enhance these features, extracting enhanced multimodal features from both PAN and MS images. Xu et al. [41] introduced a model-based deep pan-sharpening method that addresses two optimization problems regularized by deep priors. Existing DL-based methods have not considered the intrinsic relationships between source images and HRMS images. They typically rely on increasing the depth and complexity of networks, which can lead to increased computational burdens and redundancy [42]. Additionally, integrating prior knowledge has been a topic of interest. There are inherent local relationships between source images and HRMS images, which can be adaptively learned based on spectral–spatial fidelity priors [43]. Based on the above analysis, this article presents a deep prior cyclic iterative network with a dual-branch structure, resulting in high-quality fusion results. It also offers good interpretability and versatility. Experiments were conducted on the QuickBird (QB) and WorldView-2 (WV2) datasets to assess the network's performance. The main contributions of this article are as follows.

DUDB begins by incorporating a deep prior network into a neural network to reconstruct images. It then utilizes a spatially adaptive feature (SAF) adjustment mechanism to dynamically extract features. Finally, it enhances the reconstructed image further through a correlation-driven dual-branch feature fusion architecture.

1) Inspired by model-driven networks, DUDB incorporates two deep prior regularizations within the neural network to reconstruct the image, capturing more details. Given that MS and PAN images from satellite imagery contain rich spectral and spatial information, DUDB formulates two optimization problems based on deep priors. These problems can be efficiently solved using gradient projection. Inspired by algorithm unfolding, DUDB summarizes these two optimization problems into four steps and categorizes them into two modules. By alternately stacking these two modules, the algorithm unfolding inspired fusion (AUIF) module is constructed.
2) To further enhance the ability to extract fine details, DUDB introduces a detail enhancement module. This module is based on an SAF modulation mechanism that dynamically selects representative feature representations. It not only processes input features from a long-distance perspective but also effectively extracts local context information through channel mixers, enabling better extraction of relevant information for fusion.
3) To facilitate feature fusion, DUDB proposes a correlation-driven dual-branch feature fusion module (DBFM). This module uses an encoder to extract shallow features and employs long-distance attention to handle low-frequency global features. Simultaneously, it utilizes a reversible neural network to focus on extracting high-frequency local information. The module then combines global and local features through global fusion layers and local fusion layers to produce the final fused image, resulting in outstanding fusion performance.

## II. Related Work

### A. DL-Based General Sharpening Methods

Unlike traditional linear interpolation methods, DL-based approaches have the advantage of capturing more complex nonlinear mapping relationships, resulting in better restoration of high-frequency details. Additionally, they allow the entire pan-sharpening process to be modeled as an end-to-end learning problem, optimizing the process and enabling adaptive multiscale image enhancement with robust noise resistance.

In the early stages, DL-based pan-sharpening methods utilized simple three-layer networks [23]. However, these shallow network architectures could not extract deep features and did not achieve ideal fusion outcomes. Other studies adopted progressive fusion strategies to compensate for spatial details. For example, the Laplacian pyramid plane sharpening network (LPPN) [44] was designed based on the LPPN, employing a recursive structure for gradual image fusion. Subsequently, Yin et al. [45] combined image priors extracted from Gaussian and Laplacian pyramids with their proposed encoder–decoder architecture to compensate for detail loss during fusion. Yang et al. [24] introduced the PanNet method, which processes training samples in the high-pass filter domain to comprehensively extract deep features. Zheng et al. [46], aiming for comprehensive feature extraction, utilized residual structures to build networks that extracted deep spatial and spectral features, improving fusion accuracy.

The dynamic cross-channel feature fusion network (DCFNet) [47] utilizes a high-resolution branch as the main branch and supplements it with two parallel low-resolution branches progressively to enhance the main branch's information. SRPPNN [48] drew inspiration from the design

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approach in the field of image super-resolution proposed in [49]. TFNet [40] adopts a super-resolution scheme to fuse spatial and spectral features in the feature domain, yielding multiscale contextual features. Vivone et al. [41] introduced Pan-GAN, a pan-sharpening technique based on generative adversarial networks (GANs). It generates an unsupervised learning network to acquire information from both MS and PAN images and fuses them. These advancements highlight the capabilities and versatility of DL-based pan-sharpening methods, which have significantly improved image quality and generalization performance in the field.

### B. Attention Mechanism

In recent years, attention mechanisms have seen significant development and have been applied to a wide range of fields, including image processing [50], speech recognition [51], [52], and various tasks in natural language processing. Attention mechanisms emulate the way humans perceive and focus, enabling models to concentrate on important parts of information, thereby enhancing model performance and efficiency. Spatial transformer network (STN) [53] is a typical spatial attention method that can be considered as an adaptive mechanism for selecting spatial regions. Hu et al. [54] introduced the squeeze-and-excitation network (SENET), which models relationships between input image channels by analyzing them. Woo et al. [55] proposed CBAM, which uses convolutional blocks to extract key features in both channel and spatial dimensions, making it a lightweight, deployable general module. Inspired by CBAM, Jia et al. [56] added attention blocks to enhance representational power and global perception. They also designed spectral and multiscale spatial attention (SMSA) blocks. In SMSA, the participating spectral attention mechanisms further reweights spectral information along the channel dimension. Dai et al. [57] introduced the second-order attention network (SAN) for single-image super-resolution and applied the second-order channel attention (SOCA) module, which adaptively adjusts channel characteristics using second-order features. Furthermore, they designed the efficient channel attention (ECA) module [58] and lightweight enhanced spatial attention (ESA) block [59] to maintain performance while fully leveraging the benefits of attention mechanisms.

### C. Deep Unfolding Networks

Traditional iterative optimization algorithms can be effectively implemented through recursive deep neural network (DNN) module stacking [60]. Model-based DL typically models the relationships between low-resolution MS (LRMS), PAN, and high-resolution MS (HRMS) data. It solves the models using optimization algorithms and then unfolds these models. The solution steps are expanded into corresponding modules through deep unfolding techniques [61], [62], and relevant parameters are learned through end-to-end training. A pioneering work in this field is the single-level cascade network PNN proposed by [23]. The perspective of deep unfolding is inspired by the learning iterative soft thresholding algorithm (LISTA) [63], which adopts the computation procedure of traditional iterative soft thresholding algorithms to solve sparse coding problems. Therefore, LISTA can be accurately interpreted. LISTA offers an advanced deep architecture solution for learnable traditional optimization algorithms, which can be trained through supervised learning. Monga et al. [61] and Yang et al. [64] found that model-based networks are often superior to intuitively designed networks. Xie et al. [65] introduced an interpretable deep network called MHFNet, which combines the generative model of low-resolution hyperspectral images with the low-rank prior knowledge of high-resolution hyperspectral images and solves this model using an approximate gradient descent algorithm. Some research attempts to address the pan-sharpening problem using deep unfolding. For example, Tian et al. [66] constructed a prior based on the similarity between PAN and HRMS images by a nonlinear operator that can be learned through a deep network. Wu et al. [37] proposed a generic fusion framework that is able to weightedly combine variational optimization (VO) with DL for the task of pan-sharpening, where these crucial weights directly determining the relative contribution of DL to each pixel are estimated adaptively. Cao et al. [67] presented a new pan-sharpening model using the convolutional sparse coding (CSC), which is quite different from the current pan-sharpening frameworks. Yin [68] proposed a novel DNN for pan-sharpening inspired by the learned iterative soft thresholding algorithm. A coupled CSC-based pan-sharpening (PSCSC) model and related traditional optimization algorithm are proposed. A prominent advantage of deep unfolding-based approaches is the clear definition of network parameters, emphasizing interpretability.

## III. Model Construction

In this section, we will describe and explain the high-level architecture and specific algorithms of our DUDB designed for addressing the fusion and pan-sharpening problem of MS and PAN images based on the network structure depicted in Fig. 1. To ensure that the fused result image better captures the detailed characteristics of both the MS and PAN images, we have introduced two optimization problems for deep prior regularization, which are embedded separately into the two modules of the AUIF-Block.

### A. AUIF-Block

To achieve this, we upsample the LRMS image $T \in R^{h\times w\times C}$ (where $h$, $w$, and $C$ represent the height, width, and number of channels of the LRMS image) to obtain an MS image $\Gamma \in R^{H\times W\times C}$ with the same size as the PAN image $\Lambda \in R^{H\times W}$, resulting in the HRMS image $\Upsilon \in R^{H\times W\times C}$. We assume that the LRMS image is generated by blurring and downsampling the HRMS image, while the PAN image is produced from the spectral responses of the HRMS image. Therefore, we have the equations $T = BW\Upsilon$ and $\Lambda = \Upsilon H$, where $B \in R^{hw\times HW}$ represents the downsampling matrix, $W$ is the circulant convolution matrix, and $H \in R^{C\times c}$ represents the spectral response function. Since inferring HRMS is a challenging problem, penalized optimization schemes are typically

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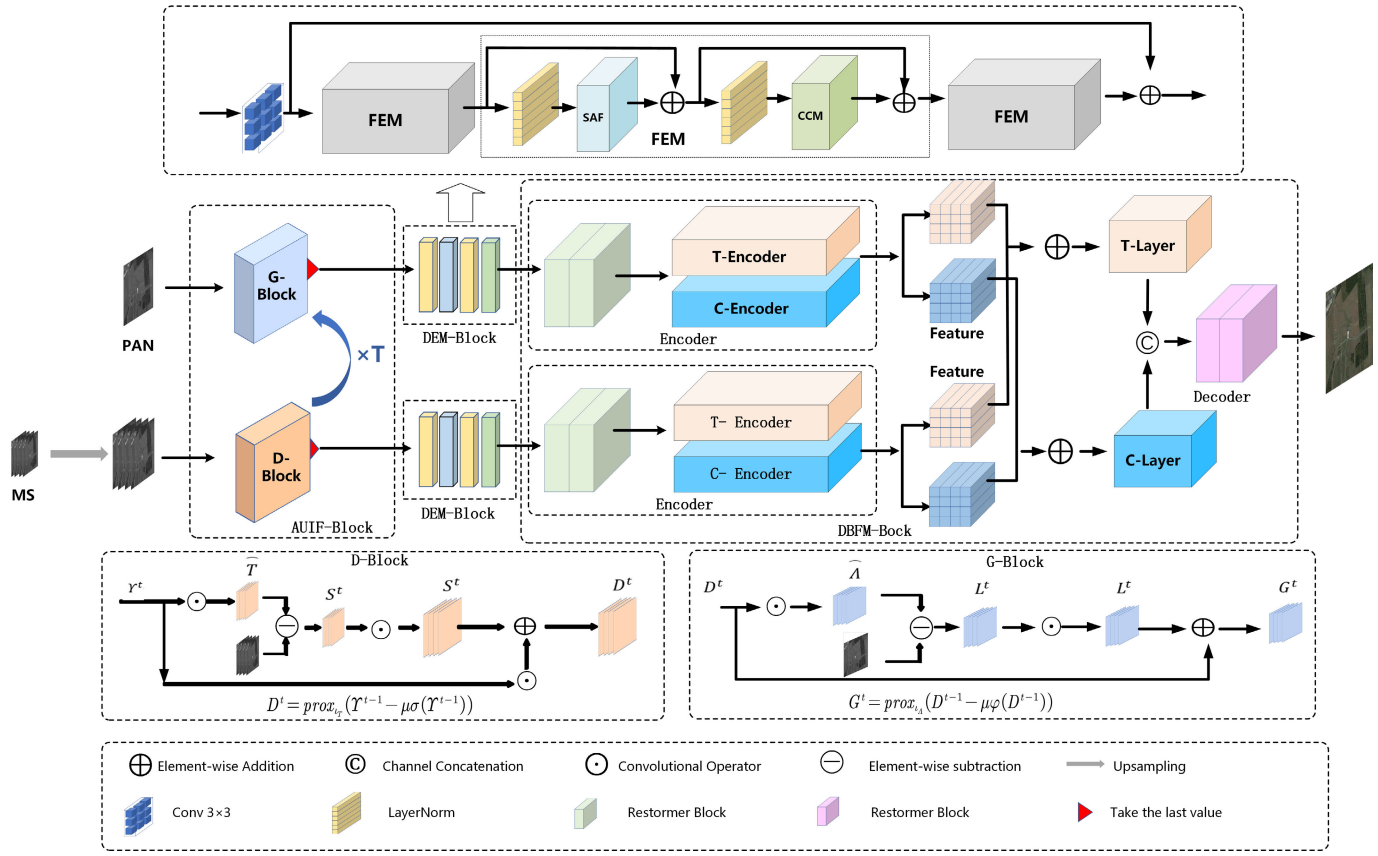



Fig. 1. Overall framework of our proposed DUDB contains three modules: AUIF-Block, DEM-Block, DBFM-Block.

proposed to tackle it

$$\min_{\Upsilon} g(T,\Upsilon)+h(\Lambda,\Upsilon)+\theta\iota_T(\Upsilon)+\theta\iota_\Lambda(\Upsilon). \tag{1}$$

In the above formulas, $g(T,\Upsilon)=(1/2)\| T-BW\Upsilon \|_2^2$ represents the data fidelity term for LRMS, and $h(\Lambda,\Upsilon)=(1/2)\| \Lambda-\Upsilon H \|_2^2$ represents the data fidelity term for the PAN image. $\iota_T(\cdot)$ and $\iota_\Lambda(\cdot)$ represent two deep priors, which are learned from the data and can adapt to different tasks and observation models. To fully utilize the deep priors, we divide them into the following two sub-equations, each sub-equation addressing the perceptual problem for LRMS images and PAN images, respectively:

$$D^t=\min_D \frac{1}{2}\| T-BWD \|_2^2+\theta\iota_T(D) \tag{2}$$

$$G^t=\min_G \frac{1}{2}\| \Lambda-GH \|_2^2+\theta\iota_\Lambda(G). \tag{3}$$

*1) D-Block:* First, we solve the above (2) using the gradient projection method, and the update rule is as follows:

$$D^t=\text{prox}_{\iota_T}\left(\Upsilon^{t-1}-\mu\sigma\left(\Upsilon^{t-1}\right)\right) \tag{4}$$

where $\mu$ is the step size, $\text{prox}_{\iota_T}(\cdot)$ is the proximal operator corresponding to $\iota_T(\cdot)$, and $\sigma(\Upsilon^{t-1})=-(BW)^T(T-BW\Upsilon)$ represents the gradient of the data fidelity term. Inspired by the principles of DL model-driven approaches, we encapsulate the above equation into a neural network block, breaking down the equation into four steps

$$\widehat{T}=BW\Upsilon^{t-1} \tag{5}$$

$$S^t=T-\widehat{T} \tag{6}$$

$$S^t=\mu(BW)^T S^t \tag{7}$$

$$D^t=\text{prox}_{\iota_T}\left(\Upsilon^{t-1}+S^t\right). \tag{8}$$

In (5), the given HRMS image $\Upsilon$ is convolved through a cyclic convolution operator and downsampled to obtain more expressive features within the network. Equation (6) represents the residual between the MS image and the extracted feature image. Equation (7) is similar to (6), with the difference being that it yields a higher-resolution residual, and its filter is transposed with respect to (6). In (8), the residual is added to the HRMS image $\Upsilon^{t-1}$ through element-wise addition, and the proximal operator of the deep prior is used to implicitly learn from the data.

*2) G-Block:* Furthermore, we analyze the observation model for PAN, and similarly, we use the gradient projection method to solve (3) above, with the update rule as follows:

$$G^t=\text{prox}_{\iota_\Lambda}\left(D^{t-1}-\mu\varphi\left(D^{t-1}\right)\right) \tag{9}$$

where $\varphi(D^{t-1})=-(\Lambda-DH)H^T$. Inspired by the principles of DL model-driven, we encapsulate the above equation in a neural network block, decomposing the equation into four steps as follows:

$$\widehat{\Lambda}=D^{t-1}H \tag{10}$$

$$L^t=\Lambda-\widehat{\Lambda} \tag{11}$$

$$L^t=\mu L^t H^T \tag{12}$$

$$G^t=\text{prox}_{\iota_\Lambda}\left(D^{t-1}+L^t\right). \tag{13}$$

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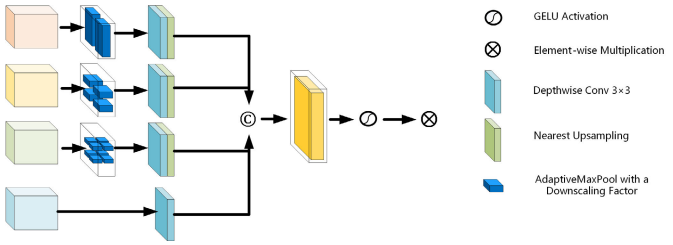



Fig. 2. SAF is the main structure of DEM-Block. Our proposed SAF has four branches for extracting spatial detail information at different scales.

In (10), it is assumed that the PAN image is a linear combination of the HRMS image, where H is a frequency-band reduction or upscaling operator. Equation (11) represents the residual between the real PAN image and the generated PAN image. Equation (12) is similar to (11), but it can obtain a higher-resolution residual, with its filter transposed in relation to (11). Equation (13) combines the residual with the HRMS image $\Upsilon^{t-1}$ through element-wise addition, implicitly learning from the data using the deep prior's proximal operator. In summary, the AUIF block in our proposed sharpening network includes iterations of the steps described above, applied to LRMS, HRMS, and PAN images within subnetworks. The implicit deep prior network is used to achieve end-to-end data learning when solving the equations. The algorithm completes two blocks of iterative optimization. The D-Block first optimizes the extracted feature image using the LRMS image $T$, and then outputs it as the reconstructed HRMS image $D^t$. The G-Block uses the updated feature image $D^t$ as input and continues to optimize it using the PAN image $\Lambda$, reconstructing it as the HRMS image $G^{t-1}$, which serves as input for the next iteration.

### B. DEM-Block

Next, let us introduce the detail enhancement module, which consists of feature extraction blocks called FEM. First, a convolution layer with a 3 × 3 kernel size is used to extract features from the reconstructed HRMS image, referred to as the shallow feature $H_0$. Then, multiple stacked FEMs are used to generate finer deep features from $H_0$ for HRMS reconstruction. The FEM contains a channel cross-frequency mixer (CCM) and incorporates an SAF modulation mechanism. We also introduce a global residual to learn high-frequency information. The steps are as follows:

$$H_0 = P_\alpha\left(\Upsilon^t\right) \tag{14}$$

$$\Upsilon^t_{H_0} = Y_\omega(H_0) + H_0. \tag{15}$$

In (14), $\Upsilon^t$ represents the input HRMS image that needs to be reconstructed, and $P_\alpha(\cdot)$ is the first convolution parameterized by $\alpha$. In (15), $Y_\omega(\cdot)$ represents the stacked FEM modules parameterized by $\omega$. Inspired by the design of the vision transformer (ViT) network, the FEM module consists of a self-attention block (SAF) for extracting global features and a feed-forward network (CCM) for feature refinement.

*1) SAF:* Compared to self-attention mechanisms, we propose a novel approach to learning multiscale features from a long-distance perspective to better reconstruct HRMS images. To generate attention maps for SAF modulation (see Fig. 2), we apply a feature pyramid. While maintaining a pyramid-shaped feature representation to reduce model complexity, we first perform channel-wise splitting on the normalized input features, resulting in four components. Among them, the first component undergoes a 3 × 3 depth-wise convolution, while the remaining parts are fed into the multiscale feature generation unit. Given a feature $\Upsilon$, this process can be represented as

$$[\Upsilon_0, \Upsilon_1, \Upsilon_2, \Upsilon_3] = S(\Upsilon) \tag{16}$$

$$\widehat{\Upsilon}_0 = DW_{3\times3}(\Upsilon_0) \tag{17}$$

$$\widehat{\Upsilon}_i = \uparrow_\eta \left(DW_{3\times3}\left(\downarrow_{\frac{\eta}{2^i}} \left(\Upsilon^i\right)\right)\right), \quad 1 \le i \le 3. \tag{18}$$

Equation (16) involves $S(\cdot)$, which represents the channel-wise splitting operation. In (17), $DW_{3\times3}(\cdot)$ represents a depth-wise convolution with a kernel size of 3 × 3 pixels. In (18), $\uparrow_\eta(\cdot)$ denotes nearest-neighbor interpolation to upsample features of different sizes to the original resolution $\eta$, while $\downarrow_{\eta/2^i}(\cdot)$ represents downsampling the input to $(\eta/2^i)$. We aim to select discriminative features with non-local interactions for learning, and we use an adaptive max-pooling layer on input features to generate multiscale features, which helps improve reconstruction performance. Then, we employ a 1 × 1 convolution to concatenate these multiscale features, aggregating local and global relationships. This process can be represented as

$$\widehat{\Upsilon} = \mathrm{Conv}_{1\times1}\left(C\left(\left[\widehat{\Upsilon}_0, \widehat{\Upsilon}_1, \widehat{\Upsilon}_2, \widehat{\Upsilon}_3\right]\right)\right). \tag{19}$$

In (19), $C(\cdot)$ represents concatenation along the channel dimension, and $\mathrm{Conv}_{1\times1}(\cdot)$ denotes a 1 × 1 convolution. After computing according to (19), we estimate the attention map through GELU for nonlinear normalization and adaptively adjust $\Upsilon$ through element-wise multiplication with the estimated attention map. This process can be represented as

$$\bar{\Upsilon} = \xi\left(\widehat{\Upsilon}\right) \odot \Upsilon. \tag{20}$$

Here, $\xi(\cdot)$ represents the GELU function, and $\odot$ denotes element-wise multiplication. Thanks to the feature pyramid structure, we can collect long-range features through this spatial adaptive modulation mechanism, enhancing the performance of reconstruction.

*2) CCM:* Compared to the SAF module that focuses on exploring global information while explicitly recognizing the importance of local contextual information for HRMS image reconstruction, inspired by common feedforward networks, we often use two consecutive 1 × 1 convolutions to transform the feature dimensions along the channel axis for exploring local context information. We propose a CCM that is based on FMBconv to enhance the ability to explore local context information. This module includes a 3 × 3 convolution and a 1 × 1 convolution. The first convolution doubles the input channel count, enabling channel mixing to encode local context information. The second convolution then restores the channel count to the original input dimensions, completing the entire process. When combined with the SAF module, it imparts dynamic effects in the channel dimension.

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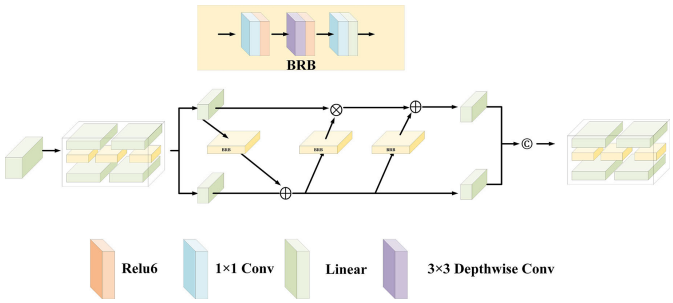



Fig. 3. Structure of INN, main structure of the C-Encoder and C-Layer. It can be defined as a lossless feature extraction module.

### C. DBFM-Block

Next, let us delve into the discussion of the DBFM. It comprises a dual-branch encoder for feature extraction, a decoder for image reconstruction, a low-frequency long-range feature fusion layer, and a high-frequency feature fusion layer. We employ DBFM to fuse features from two reconstructed HRMS images. Considering the bias introduced by fusing different spectral characteristics, we utilize the lite transformer and invertible neural network (INN) blocks as fusion layers, denoted as T-Layer and C-Layer in Fig. 1. Due to the involvement of multiple spectral features, we design separate encoders and decoders. The encoder consists of two Restomer-based shared feature blocks (SR), a basic encoder LR based on Lite Transformer, and a convolutional neural network based on INN. LR and INN collectively form the high-low frequency encoder corresponding to T-Encoder and C-Encoder in Fig. 1. The decoder includes two shared feature blocks based on Restomer. We represent SR, LR, and INN as $S(\cdot)$, $L(\cdot)$, and $I(\cdot)$, respectively. This process can be expressed as

$$\chi_1^S = S(\Upsilon_1) \quad \chi_2^S = S(\Upsilon_2) \tag{21}$$

$$\chi_1^L = L\left(\chi_1^S\right) \quad \chi_2^L = L\left(\chi_2^S\right) \tag{22}$$

$$\chi_1^I = I\left(\chi_1^S\right) \quad \chi_2^I = I\left(\chi_2^S\right). \tag{23}$$

As widely known, Restomer applies self-attention mechanisms to extract global features along the channel dimension. Additionally, to capture long-range dependencies, we use a Transformer with spatial self-attention. Furthermore, considering the importance of edge and texture information in obtaining high-frequency details while preserving as much detail as possible, we employ an INN module with affine coupling layers (see Fig. 3). The INN module can generate its input and output features from each other, which allows for better preservation of input information. It can be defined as a lossless feature extraction module. In each reversible layer, the transformation is defined as

$$X_{1,n+1}^S[c+1:C] = X_{1,n}^S[c+1:C] + H_1\left(X_{1,n}^S[1:c]\right) \tag{24}$$

$$X_{1,n+1}^S[1:c] = X_{1,n}^S[1:c] \diamond \exp\left(H_2\left(X_{1,n+1}^S[c+1:C]\right)\right) + H_3\left(X_{1,n+1}^S[c+1:C]\right) \tag{25}$$

$$X_{1,n+1}^S = \Theta\left(X_{1,n+1}^S[1:c], X_{1,n+1}^S[c+1:C]\right). \tag{26}$$

In the expression, $X_{1,n}^S[1:c] \in R^{h\times w\times c}$ is the 1st to the cth channels of input feature for the kth invertible layer ($n = 1, \ldots, N$), the $\diamond$ is the Hadamard product, and $\Theta$ represents the operation of concatenating tensors along the channel dimension. $H_i (i = 1, \ldots, 3)$ represents the bottleneck residual block (BRB). Similarly, $X_1^D = X_{1,N}^S$, $X_2^D$ can be obtained using the same procedure.

### D. Loss Function

We use a loss function to calculate the loss between the output images from DUDB and the real ground-truth images. In past image restoration tasks, it has been shown that $L_1$ loss is more robust than $L_2$ and does not cause issues like gradient explosions. Additionally, we use the Laplacian pyramid loss function $L_{\text{ap}}$ to minimize differences in color information. Therefore, to enhance the color richness of the fused images and achieve smoother edge detail, we train the network using a weighted combination of $L_1$ and $L_{\text{ap}}$ losses

$$L = L_1\left(\widehat{x}, x\right) + \kappa L_{\text{ap}}\left(\widehat{x}, x\right) \tag{27}$$

where $\widehat{x}$ corresponds to the final fused image output by DUDB, and $x$ represents the real ground truth. The $L_1$ loss in batch form is represented by the formula $L_1 = (1/n)\sum_{i=1}^{n} \| x_i - f(\widehat{x}_i) \|$, where $\|\cdot\|$ represents the $L_1$ parameterization of the sum of absolute values, and $n$ represents the number of training samples.

The $L_{\text{ap}}$ loss is formulated as follows:

$$L_{\text{ap}} = \sum_{i=1}^{n} 2^{i-1} \left\| L^i(x) - L^i\left(\widehat{x}\right) \right\|. \tag{28}$$

$L^i(x)$ represents $x$ at the $i$th level of the Laplacian pyramid, and similarly, $L^i(\widehat{x})$ represents $\widehat{x}$ at the $i$th level of the Laplacian pyramid. Both $x$ and $\widehat{x}$ undergo Laplacian pyramid operations. This loss ultimately takes the sum of the differences between the $n$ layers of the Laplacian pyramids. The weighted combination of $L_1$ and $L_{\text{ap}}$ results in better quantitative results.

To sum up, the procedure of our proposed DUDB can be summarized as Algorithm 1.

**Algorithm 1** DUDB: Deep Unfolding-Based Dual-Branch Feature Fusion Network for Pan-Sharpening Remote Sensing Images

**Require:** The upsampled MS image $\Gamma$. The PAN image $\Lambda$
**Input:** MS and PAN images: $\Gamma$ and $\Lambda$
**Output:** High-resolution multispectral images: H
**for:** t=1:8 do
  $D^t \longleftarrow D - Block(G^{t-1}, \Gamma)$
  $G^t \longleftarrow G - Block(D^{t-1}, \Lambda)$
**end for.** return $D^T$, $G^T$
$M \longleftarrow DEM - Block(D^T)$
$N \longleftarrow DEM - Block(G^T)$
$H \longleftarrow DBFM - Block(M, N)$
return fusion result image: H

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## IV. Experiment and Evaluations

In this section, we provide a detailed description of the experimental details and evaluation metrics for the DUDB network, which is designed to address pan-sharpening problems using the AUIF-Block, DEM-Block, and DBFM-Block. First, we introduce the datasets and evaluation metrics used in the experiments. Next, we compare DUDB with seven other pan-sharpening methods, including both simulated and real experiments, to demonstrate the superiority of our proposed DUDB. We conduct qualitative and quantitative analyses to assess the performance of DUDB. Additionally, we perform ablation experiments to evaluate the effectiveness of each module within DUDB. These experiments were conducted using PyTorch on an NVIDIA Titan RTX computer.

### A. Datasets and Metrics

We conducted comparative experiments on the QB and WV2 datasets and evaluated our proposed DUDB. The QB dataset and WV2 dataset both contain four bands: red, green, blue, and near-infrared. The QB dataset provides an MS image with a spatial resolution of 2.44 m and a PAN image with a spatial resolution of 0.61 m. The WV2 dataset offers an MS image with a spatial resolution of 2 m and a PAN image with a spatial resolution of 0.5 m. During the training phase, we set the number of epochs to 100, batch size to 4, and the initial learning rate to 0.0001. The learning rate is adaptively adjusted using the Adam optimizer to ensure training stability. We use two main approaches to evaluate the experimental results: qualitative analysis and quantitative comparison. On one hand, we visualize the fused images for qualitative analysis to make it easy to visually assess the clarity of the images. On the other hand, we calculate several commonly used reference and no-reference evaluation metrics for objective evaluation of the fusion performance. The reference metrics include spectral angle mapper (SAM) [69], spatial correlation coefficient (SCC) [69], global relative error (ERGAS) [70], and the multiband extension of the universal image quality index ($Q2^n$) [71]. The no-reference metrics include $D_s$, $D_\lambda$ and QNR. Let us go into more detail about these metrics. We use two main approaches to evaluate the experimental results: qualitative analysis and quantitative comparison. On one hand, we visualize the fused images for qualitative analysis to make it easy to visually assess the clarity of the images. On the other hand, we calculate several commonly used reference and no-reference evaluation metrics for objective evaluation of the fusion performance. The reference metrics include SAM, SCC, ERGAS, and the multiband extension of the universal image quality index ($Q2^n$). The no-reference metrics include $D_s$, $D_\lambda$ and QNR. Let us go into more detail about these metrics.

1) *SAM:* SAM calculates the angles between each spectral band in the fused image and the reference image. It quantifies spectral distortion, with lower SAM values indicating less spectral distortion. The ideal value is 0.
2) $Q2^n$*:* $Q2^n$ is based on the computation of the hypercomplex correlation coefficient between the reference and tested images, which jointly measures spectral and spatial distortions. A higher $Q2^n$ value indicates better fusion quality, with 1 being the ideal value.
3) *ERGAS:* ERGAS is used to calculate global distortion using the root-mean-square error (RMSE) [64]. Smaller ERGAS values indicate better fusion performance, meaning less global distortion.
4) *QNR:* QNR assesses both spatial and spectral distortions between the fused result and the source image. It consists of two components: $D_s$, representing spatial distortion, and $D_\lambda$, representing spectral distortion. Higher QNR values indicate better fusion, while the optimal values for $D_s$ and $D_\lambda$ are 0.

### B. Comparative Experiment

In the comparative experiments, we compared our proposed method with seven advanced pan-sharpening methods on both simulated and real data, including qualitative analysis and quantitative estimation. These seven methods consist of traditional approaches: GFPCA [27], MTF_GLP_HPM [41], as well as DL-based methods: PANNET [23], AWFLN [43], TFNET [40], GPPNN [72], and DMFA [73]. The source code for the above methods is open-source, and the parameters are configured based on the original papers.

*1) Simulation Experiment:* According to the WALD protocol [74], we conducted experiments on the QB and WV2 datasets using HRMS as the ground-truth reference image to perform qualitative analysis and quantitative estimation of the seven methods. Let us start with the qualitative analysis of these seven deblurring methods.

Figs. 4 and 5 display the result images of all compared methods on the QB simulated dataset. Fig. 4 presents zoomed-in images of details, while Fig. 5 shows the residual images between the result images on the QB simulated dataset and the ground-truth images. In Fig. 4, we can observe ground-level road details and parked car images. The yellow box in the zoomed-in details reveals a forked intersection and fine edge textures. The blue box illustrates an enlarged view of the arrangement of parked cars along the roadside. For the parked car information, we primarily focus on three sets of adjacent parked white cars. Upon observation, it is clear that the TFNET fused image does not provide us with effective information. While it retains spectral information relatively well compared to the ground-truth image, it lacks spatial information and we cannot observe the white cars. Additionally, in the images fused by PANNET and DMFA, the boundary information of the cars within the blue box is blurry, lacking spatial texture information. Among these methods, DMFA shows slightly clearer texture details in the yellow box, where several parallel adjacent roads are present. In contrast, in the images fused by GFPCA, MTF_GLP_HPM, and PANNET, the ground within the blue box tends to be whitish, indicating significant spectral information loss. The image fused by GPPNN displays clearer edge line textures of white cars within the blue box compared to the other methods. It also recovers the line details of parallel adjacent roads in the yellow box better than the other methods. However, in terms of spectral information, there is a color difference in the vegetation color on one side of the forked intersection within the yellow box, leaning toward grayish tones. AWFLN achieves better restoration of spectral information than other methods, with the spectral information

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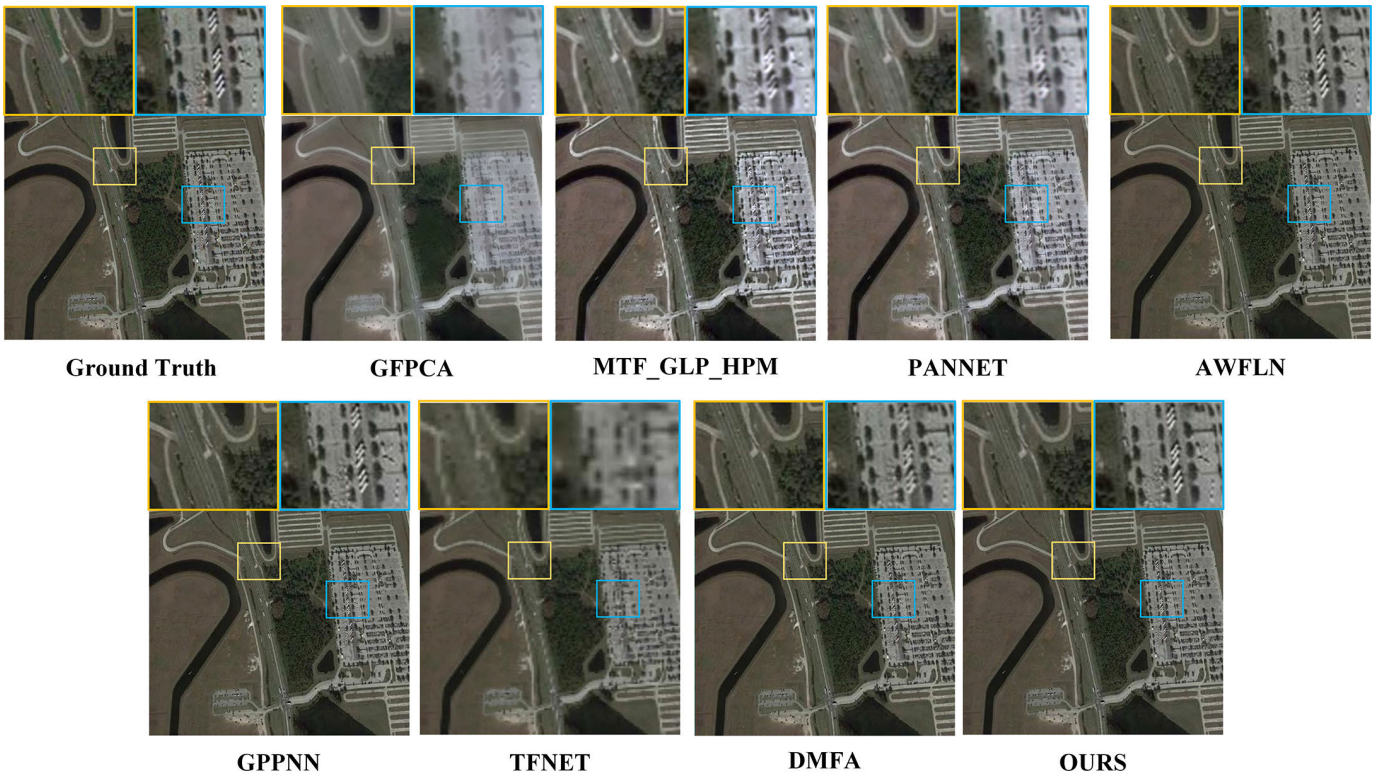



Fig. 4. Fusion results of seven pan-sharpening methods on the QB simulation dataset. The blue framed image and the yellow framed image above the image are detailed enlarged images. You can zoom in to view image details.

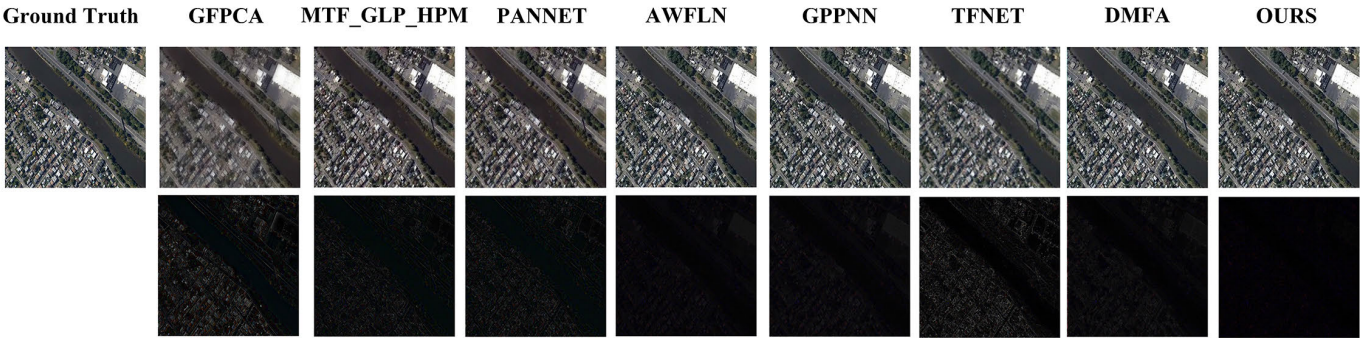



Fig. 5. Fusion results of seven pan-sharpening methods on the QB simulation dataset and residual images with ground-truth images. The images can be zoomed-in to see the details.

in the fused image closely resembling the ground-truth image. Nevertheless, as mentioned earlier, it suffers from a lack of spatial texture information. Our method preserves complete information about the arrangement of white cars, with visible edge textures of cars and line textures of parallel adjacent roads in the yellow box. Moreover, the ground color of the white cars is accurately represented within the blue box, and there is no distortion in the vegetation color within the yellow box. In comparison to the other seven methods, our method exhibits the best performance in terms of both spectral and spatial information restoration.

Fig. 5 displays the fused images and their respective residual images. The TFNET residual image exhibits the most prominent spatial details, indicating significant spatial detail loss. In contrast, the residual images of GFPCA, MTF_GLP_HPM, and PANNET methods show slight spatial distortions but still retain fundamental spatial details. AWFLN produces a residual image with slight light spots and some loss of spectral information, but the spatial information is retained intact. The residual images produced by GPPNN and DMFA methods appear in black and white, suggesting less loss of spectral information and better preservation of spectral details. Our method exhibits negligible distortion in both spectral and spatial information in the residual images. It achieves more complete spatial information recovery and uniform restoration of spectral information.

Figs. 6 and 7 display the results of all methods on the simulated dataset WV2. Fig. 6 presents enlarged detail images, while Fig. 7 shows the result images of the WV2 simulated dataset and the residual compared to the ground-truth image. As shown in Fig. 6, various details such as houses, streets, and a densely packed parking lot with vehicles are visible. The yellow box zooms in on three rows of residential houses and green vegetation. The blue box enlarges the parking lot area. In the yellow box, our main focus is on the details of the three rows of houses. According to our observations, TFNET fails

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Fig. 6. Fusion results of seven pan-sharpening methods on the WV2 simulation dataset. The blue and yellow borders above the image are zoomed-in images of details. You can zoom in to view image details.

to provide useful information. While it roughly reconstructs spectral information, the spatial details are severely lacking, and house textures cannot be discerned. In contrast, the images produced by the GFPCA and MTF_GLP_HPM methods show darker house colors with less noticeable contrast, indicating spectral information bias. Also, when observing the blue box, while we can see densely parked vehicles and ground markings, they are somewhat blurry compared to the ground-truth image, indicating the need for more spatial details. Regarding the fusion of PANNET, GPPNN, and AWFLN although they preserve spectral information well, there are some missing house details and textures, as well as shadows in the yellow box. Additionally, when comparing the blue box, PANNET exhibits a loss of focus in parking lot details. In the case of DMFA, there is no noticeable difference in house colors compared to the ground truth, and spectral information is well restored. However, when observing the blue box image, there are some missing details in the parking lot landmarks. In our method, the outlines of houses, parked vehicles, and landmarks are all clearly visible. House colors are well restored with significant contrast. Therefore, it can be concluded that our method provides the best preservation of spatial details and spectral information in the generated images.

Fig. 7 displays the fused images and residual images. Comparing the residual images, TFNET's residual image is the clearest with the brightest colors, indicating that TFNET has the most severe loss of spatial and spectral information. The residual image of the PANNET method shows clear contour details in the lower left corner, indicating a significant loss of spatial details. Upon closer inspection of the residual images, the GFPCA and MTF_GLP_HPM methods lose spatial details of dense building outlines and exhibit some spectral distortion. GPPNN and AWFLN preserve spatial information better than other methods, with more accurate road and building lines, although some areas appear brighter when magnified, indicating some loss of spectral information. DMFA fused images have a slight distortion of spatial and spectral information. By observing the residual image of our method compared to the ground-truth image, it is evident that our method effectively preserves spatial details and successfully restores spectral information.

In addition, we conducted quantitative comparisons on two simulated datasets, analyzing the fusion results of all methods. We used four reference evaluation metrics for quantitative comparison, where the ideal values for SCC and $Q2^n$ are 1, and for ERGAS and SAM, the ideal values are 0. In Tables I and II, the best values are highlighted in bold. Table I displays the quantitative comparison of the seven methods on the QB dataset. Our method ranks second in the SAM metric, only slightly behind TFNET, but it outperforms all other methods in the SCC, $Q2^n$, and ERGAS reference metrics. Compared to the second-ranked method, our method has a lower ERGAS value than GPPNN by 0.0147, a lower SAM value of 0.0044, a higher SCC value of 0.0033, and a higher $Q2^n$ value of 0.0029. This indicates that our method exhibits the best overall performance in image restoration. Table II displays the quantitative comparison of the seven methods on the WV2 dataset. Similarly, our method outperforms all others in the SCC, $Q2^n$, and ERGAS metrics, with the SAM metric being the only exception where it is slightly behind TFNET,

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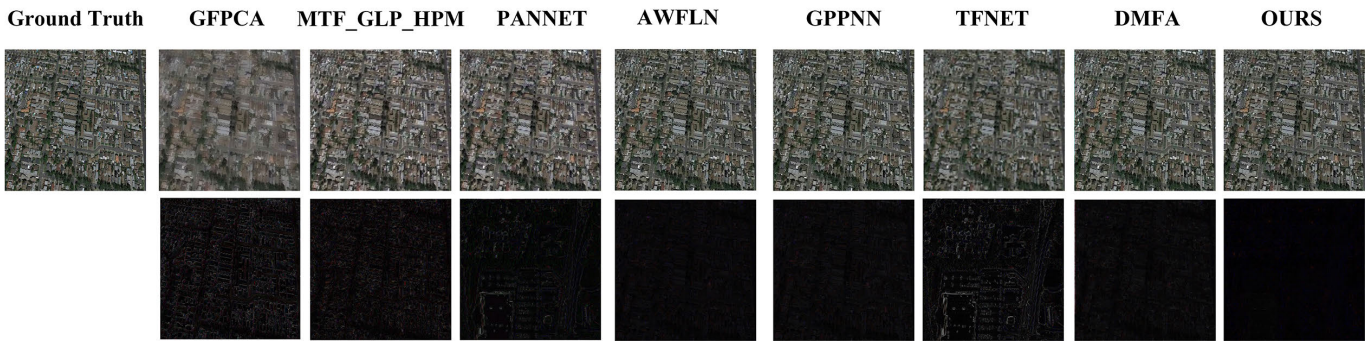



Fig. 7. Fusion results of seven pan-sharpening methods on the WV2 simulation dataset and residual images with ground-truth images. The images can be zoomed in to see the details.

TABLE I
QUANTITATIVE COMPARISON OF SEVEN METHODS ON THE QB DATASET

| Method | QB | | | |
|---|---|---|---|---|
| | SAM(0) | ERGAS(0) | SCC (1) | $Q2^n$ (1) |
| GFPCA | 0.1218 | 7.4823 | 0.8897 | 0.6093 |
| MTF_GLP_HPM | 0.1100 | 3.2143 | 0.9464 | 0.8591 |
| PanNet | 0.0922 | 2.9775 | 0.9585 | 0.8632 |
| AWFLN | 0.0403 | 1.2515 | 0.9861 | 0.9667 |
| GPPNN | 0.0402 | 1.2052 | 0.9902 | 0.9639 |
| TFNet | **0.0355** | 4.5903 | 0.9401 | 0.4101 |
| DMFA | 0.0454 | 2.0109 | 0.9778 | 0.9635 |
| Ours | 0.0399 | **1.1905** | **0.9935** | **0.9668** |

Bold indicates the best result.

TABLE II
QUANTITATIVE COMPARISON OF SEVEN METHODS ON THE WV2 DATASET

| Method | WV2 | | | |
|---|---|---|---|---|
| | SAM(0) | ERGAS(0) | SCC (1) | $Q2^n$ (1) |
| GFPCA | 0.1528 | 8.7740 | 0.8938 | 0.5730 |
| MTF_GLP_HPM | 0.1091 | 4.4770 | 0.9448 | 0.9092 |
| PanNet | 0.1329 | 4.9368 | 0.9564 | 0.9511 |
| AWFLN | 0.0284 | 0.9311 | 0.9958 | 0.9730 |
| GPPNN | 0.0311 | 1.0035 | 0.9951 | 0.9691 |
| TFNet | **0.0125** | 1.5750 | 0.9601 | 0.5045 |
| DMFA | 0.0365 | 1.9649 | 0.9815 | 0.9393 |
| Ours | 0.0178 | **0.7210** | **0.9986** | **0.9816** |

Bold indicates the best result.

but only by 0.0053. Compared to AWFLN, our method has a higher SCC value of 0.0028, a higher $Q2^n$ value of 0.0086, a lower SAM value of 0.0106, and a lower ERGAS value of 0.2101. SAM is an index for evaluating spectral information preservation, ERGAS assesses image distortion, SCC evaluates spatial information recovery, and $Q2^n$ is a comprehensive index that assesses contrast similarity, spatial information, and luminance. Thus, our method demonstrates the best overall performance, both in qualitative analysis and quantitative comparisons, highlighting the excellent preservation of spatial and spectral information achieved by DUDB, as compared to the other methods.

*2) Real Experiment:* We conducted qualitative analysis and quantitative comparisons of the fusion results. We performed qualitative analysis and quantitative comparisons on seven pan-sharpening methods using the QB and WV2 real datasets. Figs. 8 and 9 display the fusion results of the seven pan-sharpening methods on the QB real dataset. Fig. 8 shows the details enlarged, and Fig. 9 shows the fusion result images along with the residual images with the PAN images. Fig. 8 mainly includes dense housing and road information. MS images can provide rich spectral information but lack effective spatial information. PAN images can provide rich spatial information but lack spectral information. By observing the seven pan-sharpening methods, they can all restore basic spectral information. TFNET provides information similar to that of MS, mainly containing spectral information and lacking spatial information. The interlaced housing information in the yellow and blue boxes is almost indistinguishable. GFPCA, MTF_GLP_HPM, and PANNET can preserve both spatial and spectral information quite well, but in the yellow box, the roof color is slightly lighter, and there is a noticeable grayish hue in the ground color, indicating some deviation in recovering spectral information. Spectral and spatial information of the fused images of the AWFLN method are more completely retained. GPPNN and DMFA methods perform excellently in restoring both spatial and spectral information. However, DMFA does not have clear lines for dense housing in the blue box. In contrast, our method can recover finer spatial details and richer spectral information.

Fig. 9 shows the fusion images and residual images of the seven methods. Among them, the residual image of TFNET is the most pronounced, indicating that the fused image is the blurriest with severe loss of spatial information. In comparison, GFPCA, MTF_GLP_HPM, and PANNET methods are slightly smoother than TFNET but still lose some spatial information. They also appear slightly rougher compared to AWFLN, GPPNN, and DMFA. This indicates that AWFLN, GPPNN, and DMFA can preserve more spatial information, including fine housing lines. Our method, on the other hand, is overall smoother, with a black-and-white color tone and even distribution. This suggests that our fused images contain the richest spatial information, and the spectral information is also the most uniform.

Figs. 10 and 11 display the fusion results of the seven super-resolution methods on the WV2 real dataset. Fig. 10 provides detailed enlarged images, while Fig. 11 shows the fusion results and the residual images when compared to the

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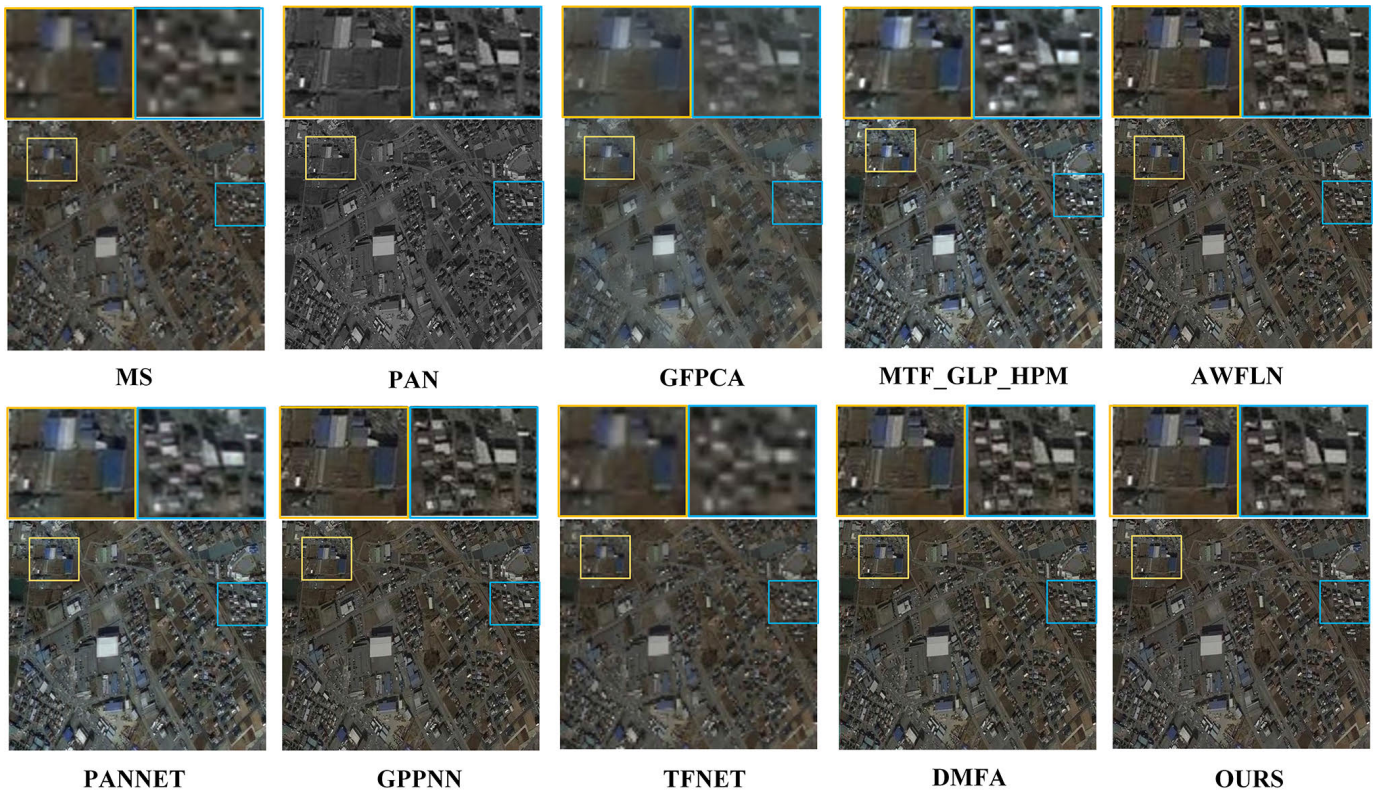



Fig. 8. Fusion results of seven pan-sharpening methods on the QB real dataset. The blue border map and the yellow border above the image zoom in on the map detail. You can zoom in to view image details.

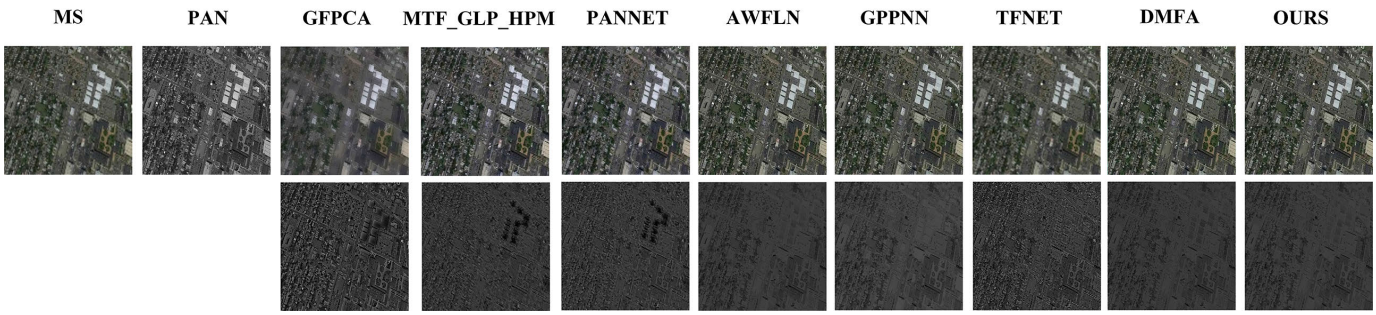



Fig. 9. Fusion results of seven pan-sharpening methods on the QB real dataset and residual images with PAN images. You can zoom in to view image details.

PAN image. Fig. 10 includes information about houses, green vegetation, and roads. MS images can provide rich spectral information but lack effective spatial information. PAN images contain ample spatial information but do not include spectral information. Observing the seven methods, except for TFNET, GPPNN, AWFLN, DMFA, and our method, others suffer from spectral information loss, and the contrast of house colors within the yellow box is not prominent. GFPCA and MTF_GLP_HPM perform well in restoring spatial information, but when observing the yellow box, the cars parked beside the houses are blurred. The DMFA method can be seen to have incomplete shapes in the blue box area, which represents the fields. Our method effectively restores spectral information, including vivid house colors with clear contrast, facilitating the identification of vehicles near the house edges, while preserving complete spatial information.

Fig. 11 displays the fusion images and residual images of the seven methods. Among them, TFNET suffers the most severe loss of spatial information. GFPCA's residual images have darker colors, indicating weaker spectral information restoration capability. PANNET's residual images are also quite rough, only slightly smoother than TFNET, and perform poorly in terms of spatial information recovery. In comparison, GFPCA and MTF_GLP_HPM methods are slightly smoother than PANNET but still lose some spatial information. On the other hand, compared to AWFLN, GPPNN, and DMFA, they appear slightly rough. This suggests that AWFLN, GPPNN, and DMFA can preserve more spatial information. Our method is generally smooth, with minimal spatial information distortion and noticeable color saturation.

Additionally, we conducted quantitative comparisons of the seven pan-sharpening methods as shown in Table III. In the

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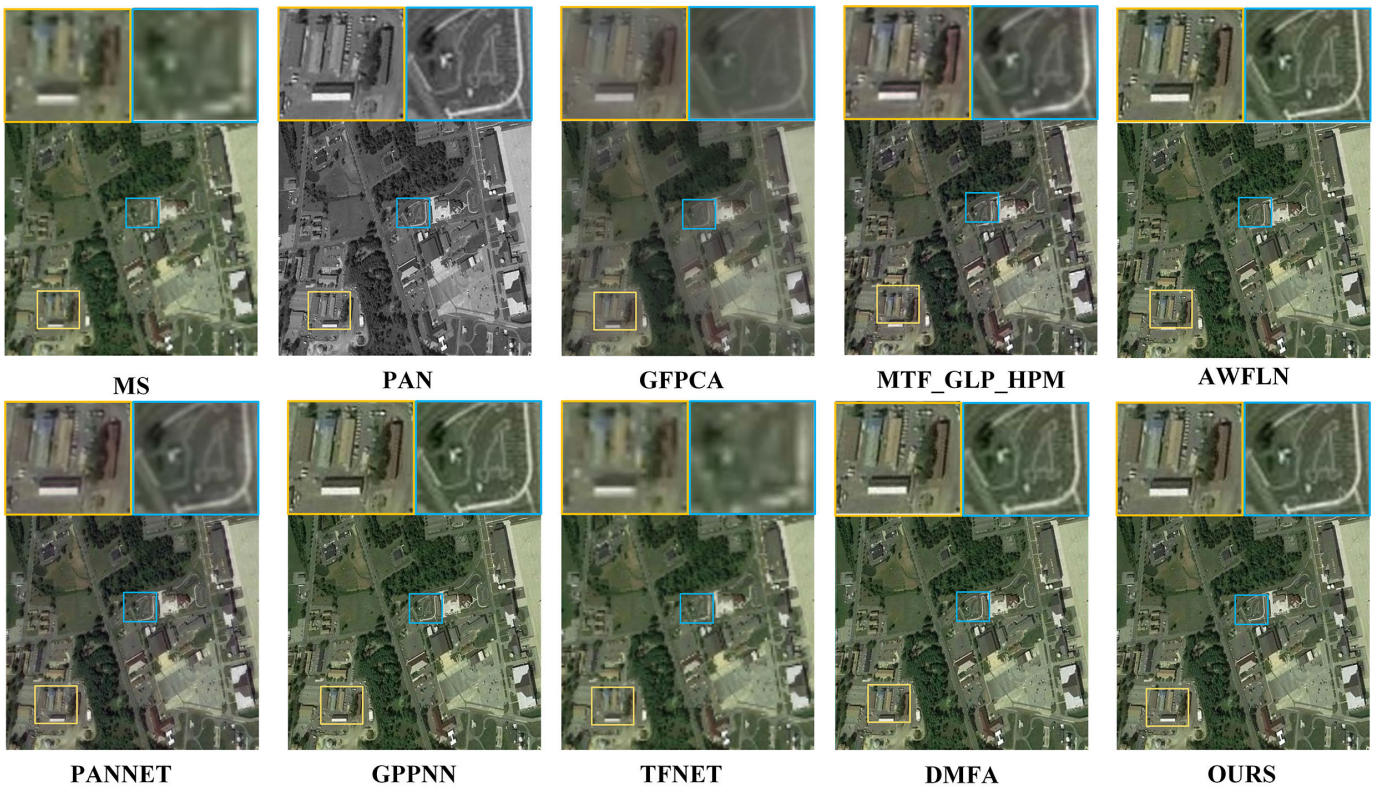



Fig. 10. Fusion results of seven pan-sharpening methods on the WV2 real dataset. The blue and yellow borders above the image are zoomed-in images of details. You can zoom in to view image details.

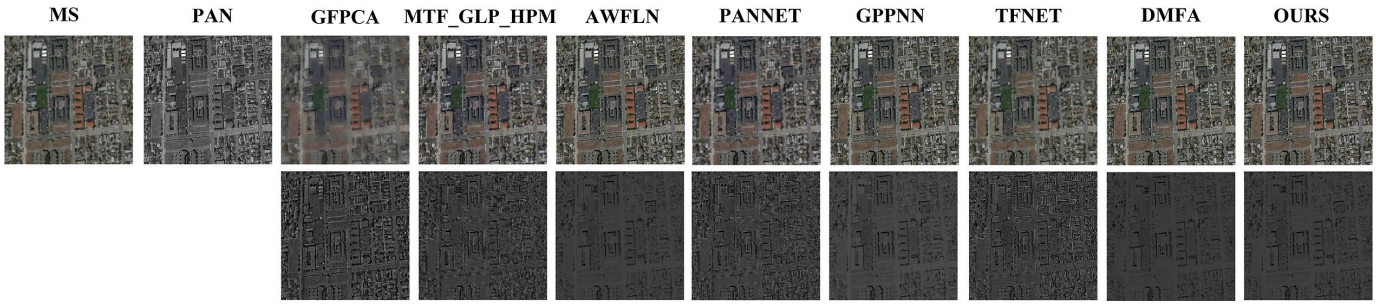



Fig. 11. Fusion results of seven pan-sharpening methods on WV2 real dataset and residual image with PAN image. You can zoom in to view image details.

TABLE III
QUANTITATIVE COMPARISON OF SEVEN METHODS ON TWO REAL DATASETS QB AND WV2

| Method | QB | | | WV2 | | |
|---|---|---|---|---|---|---|
| | $D_S \downarrow$ | $D_\lambda \downarrow$ | QNR $\uparrow$ | $D_S \downarrow$ | $D_\lambda \downarrow$ | QNR $\uparrow$ |
| GFPCA | 0.1009 | 0.0219 | 0.8797 | 0.1090 | 0.0247 | 0.8696 |
| MTF_GLP_HPM | 0.0252 | 0.0109 | 0.9642 | 0.0358 | 0.0214 | 0.9436 |
| PanNet | 0.0211 | 0.023 | 0.9563 | 0.0319 | 0.0279 | 0.9411 |
| AWFLN | 0.0314 | 0.0221 | 0.9472 | 0.0494 | 0.0313 | 0.9211 |
| GPPNN | 0.0307 | 0.0216 | 0.9484 | 0.0489 | 0.032 | 0.921 |
| TFNet | 0.0309 | 0.1414 | 0.831 | 0.0197 | 0.7693 | 0.2298 |
| DMFA | 0.0216 | 0.0229 | 0.9463 | 0.0232 | **0.0205** | 0.9443 |
| Ours | **0.0191** | **0.0097** | **0.9723** | **0.0121** | 0.0392 | **0.9477** |

Bold indicates the best result.

quantitative comparison, we used three no-reference metrics: $D_s$, $D_\lambda$, and QNR. The ideal values for $D_s$ and $D_\lambda$ are both 0, while the ideal value for QNR is 1. $D_s$ assesses the level of spatial distortion in the image, $D_\lambda$ evaluates spectral distortion, and QNR indicates the degree of spatial and spectral distortion between the fusion result and the source image. For

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TABLE IV
AVERAGE RUNNING TIME (S) AND NUMBER OF PARAMETERS (M)

| Moldes | PanNet | AWFLN | GPPNN | TFNet | DMFA | Ours |
|---|---|---|---|---|---|---|
| Params | 0.30 | 0.61 | 0.34 | 9.01 | 0.59 | 0.53 |
| Times | 0.12 | 0.32 | 0.13 | 0.79 | 0.29 | 0.23 |

$D_s$ and QNR, our method achieves the best values on both datasets, indicating that our approach effectively preserves spatial information and minimizes the distortion between the fusion result and the source image. Regarding $D_\lambda$, on the QB dataset, our method achieves the best value, while on the WV2 dataset, our method surpasses the best-performing DMFA by 0.0187. In summary, our method outperforms the other seven methods, demonstrating the best overall performance in preserving both spectral and spatial information. The number of parameters as shown in Table IV. To sum up, through qualitative analysis and quantitative comparison in both simulated and real experiments, our method demonstrates superior image restoration capabilities.

## V. CONCLUSION

In this article, we introduced a novel pan-sharpening method called DUDB. Our approach is based on a dual-branch deep unfolding iterative network for remote sensing image fusion. It consists of the AUIF module for deep prior, the DEM module for detail enhancement, and the DBFM module for feature fusion. The AUIF module combines model-driven and deep prior methods to obtain high-quality reconstructed images. The DEM module injects multilevel details into the fused image based on spatial adaptive adjustment. The DBFM module employs a dual-branch structure to fuse multidomain detail information for superior fusion results. We demonstrated the effectiveness of our proposed method through ablation experiments, and we performed qualitative and quantitative evaluations on QB and WV2 datasets against classical pan-sharpening methods in both simulated and real experiments, highlighting the superiority of our method. The experimental results indicate that the fused images exhibit complete spatial information and uniform spectral information.

## APPENDIX

We conducted ablation experiments on the DUDB method using the QB dataset, and we employed two types of ablation experiments to validate the effectiveness of our network. The first type focused on the iteration number in our proposed DUDB iterative network, as shown in Table V. The second type involved ablating each component of the network based on the optimal iteration number selected for the network. For the second ablation experiment, we designed three ablation models.

1) *DUDB-AUIF:* The DUDB method omits the extended AUIF module (1) to validate whether having two deep priors is necessary.
2) *DUDB-DEM:* The DUDB method omits the DEM module to validate the effectiveness of DEM in extracting details from multiple domains.

TABLE V
ON THE QB DATASET, DIFFERENT ITERATIONS OF AUIF ARE QUANTITATIVELY COMPARED, AND THE OPTIMAL INDEX HAS BEEN BOLDED

| Iterate | QB | | | |
|---|---|---|---|---|
| | SAM(0) | ERGAS(0) | SCC (1) | $Q2^n$ (1) |
| Iter1 | 0.042 | 1.2378 | 0.9931 | 0.9653 |
| Iter2 | 0.0418 | 1.2207 | 0.9931 | 0.9655 |
| Iter3 | 0.0412 | 1.2186 | 0.9932 | 0.9659 |
| Iter4 | 0.0408 | 1.2098 | 0.9932 | 0.9661 |
| Iter5 | 0.0405 | 1.2032 | 0.9933 | 0.9663 |
| Iter6 | 0.0405 | 1.2004 | 0.9933 | 0.9663 |
| Iter7 | 0.0404 | 1.1997 | 0.9933 | 0.9665 |
| Iter8 | **0.0399** | **1.1905** | **0.9935** | **0.9668** |
| Iter9 | 0.0411 | 1.2166 | 0.9935 | 0.9656 |
| Iter10 | 0.0422 | 1.218 | 0.9932 | 0.9655 |

Bold indicates the best result.

3) *DUDB-DBFM:* The DUDB method omits the DBFM module to validate the effectiveness of the dual-branch fusion module in producing fusion images from different domains.

In the first ablation experiment, DUDB is evaluated on the QB dataset with different numbers of iterations to determine the optimal fusion performance. Table V demonstrates that within the first eight iterations, the four metrics exhibit an increasing trend and reach their peak values at the eighth iteration, with a slight decrease in the ninth and tenth iterations. Moreover, the eighth iteration performs better than the ninth iteration overall. The SCC at the eighth iteration is 0.0002 higher than that at the seventh iteration and the same as the ninth iteration. The $Q2^n$ value at the eighth iteration is 0.0003 higher than that at the seventh iteration and 0.0012 higher than that at the ninth iteration. The SAM value at the eighth iteration is 0.0005 lower than that at the seventh iteration and 0.0012 lower than that at the ninth iteration. The ERGAS value at the eighth iteration is 0.0092 lower than that at the seventh iteration and 0.0261 lower than that at the ninth iteration. Therefore, the eighth iteration provides the best fusion results in this experiment. Based on the results of the first ablation experiment, the optimal number of iterations is determined to be 8. Consequently, we proceed to the second ablation experiment. In the second ablation experiment on the QB dataset, we evaluate the three proposed ablation models to verify the effectiveness of the proposed framework. Fig. 12 shows the fusion plot for each iteration number. Fig. 13 displays the fusion images and their corresponding residual images for the four ablation models. By comparing the fusion images, the DUDB-AUIF fusion results exhibit a yellowish color cast. Observing the corresponding residual images, the images appear relatively smooth, but upon closer inspection, there are issues such as the loss of spatial details and uneven spectral distribution. This suggests the necessity of deep prior information provided by the DUDB-AUIF module in image reconstruction. The DUDB-DEM residual images are the most pronounced, indicating the most significant spatial loss. This demonstrates the effectiveness of DUDB-DEM in extracting details across multiple domains. The DUDB-DBFM results show images that are very similar to the ground truth, but compared to

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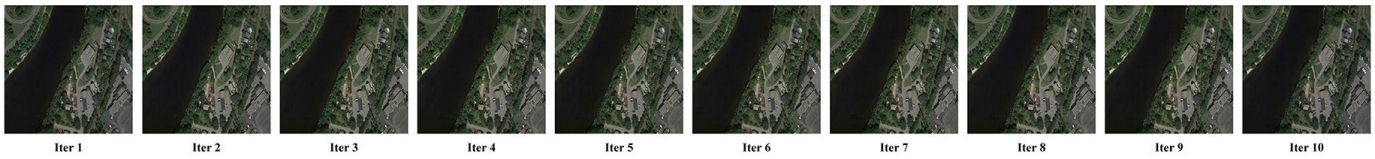



Fig. 12. Visualization of melting images with depth-expanded iteration counts.

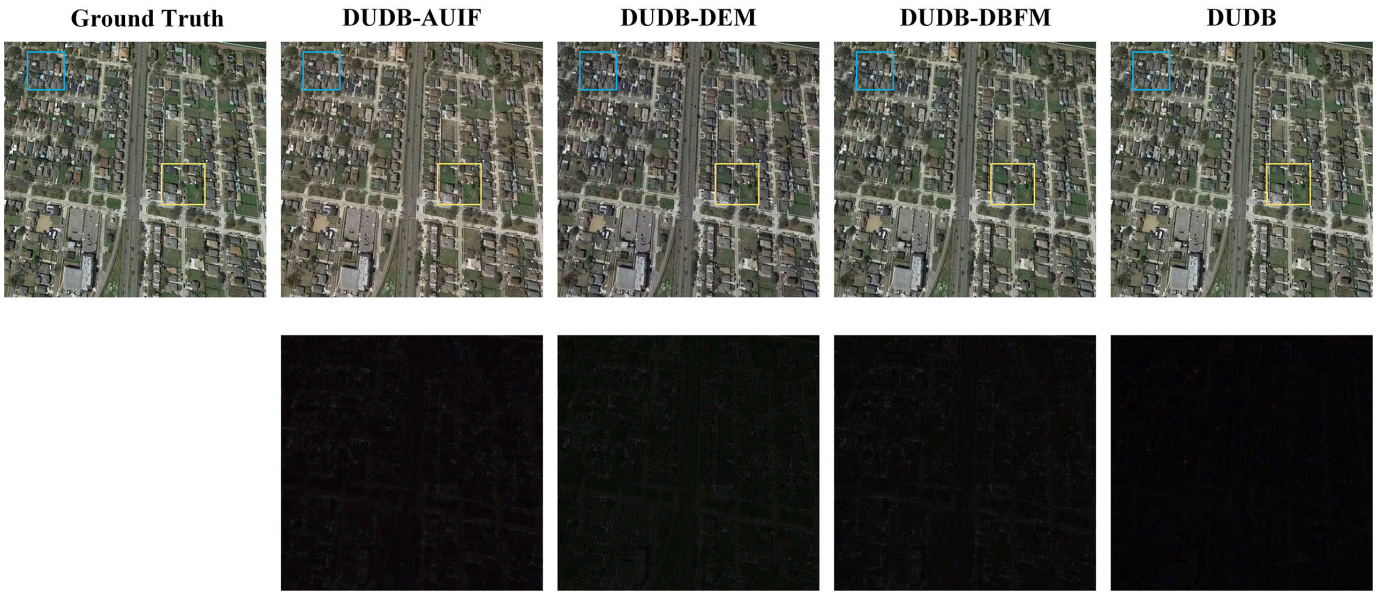



Fig. 13. Second type of ablation experiment. The first row is the fusion result image of the ground-truth image and the four models. The second row is the resulting map of the four models and the residual result map of the ground-truth image. Zoom in to view image details.




Fig. 14. Enlarged view of the details of the fused image in Fig. 13. Zoom in to view image details.

the net's residual images, they are still relatively rough. This suggests that the DBFM module introduces spatial and spectral distortions during the reconstruction process, confirming the necessity of DUDB-DBFM for fusing images at different frequencies.

Furthermore, by observing the zoomed-in details in Fig. 14, let us first focus on the densely populated house contours within the blue boxes. DUDB is the only model capable of restoring house contours effectively. In contrast, the other three models produce relatively blurry results, indicating a loss of

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TABLE VI
CONDUCTING ABLATION EXPERIMENTS ON THE QB DATASET, AND THE OPTIMAL INDEX HAS BEEN BOLDED

| Network | SAM(0) | ERGAS(0) | SCC (1) | $Q2^n$ (1) |
|---|---|---|---|---|
| **DUDB-AUIF** | 0.0412 | 1.2164 | 0.9933 | 0.9657 |
| **DUDB-DEM** | 0.0418 | 1.2077 | 0.9932 | 0.9658 |
| **DUDB-DBFM** | 0.0408 | 1.2071 | 0.9934 | 0.9662 |
| **DUDB** | **0.0399** | **1.1905** | **0.9935** | **0.9668** |

Bold indicates the best result.

spatial details in all four models, with DEM showing the most significant loss of details.

Observing the green vegetation in the yellow boxes, DUDB is the only model that successfully restores the colors of the green vegetation, closely matching the ground truth. The other three models exhibit spectral distortions, further affirming the necessity of each module in the proposed framework.

Table VI presents our quantitative comparison of the four ablation models. The best values are indicated in bold. Based on the four metrics in the table, it can be observed that DUDB performs the best in terms of fusion effectiveness, preserving both spectral and spatial information to the greatest extent. Among the four metrics, DUDB-AUIF and DUDB have the largest differences. SAM increases by 0.0013, ERGAS increases by 0.0259, SCC decreases by 0.0002, and Q decreases by 0.0011 compared to DUDB. This indicates that DUDB-AUIF has good feature extraction and fusion capabilities for image reconstruction. The difference between DUDB-DEM and DUDB in $Q2^n$ value is greater than that between DUDB-DBFM and DUDB. $Q2^n$ value decreases by 0.0010, and ERGAS increases by 0.0172. The other two values also differ significantly, highlighting that DUDB-DEM can effectively extract features from multiple domains, reducing image distortion effectively. In comparison to DUDB, DUDB-DBFM results in an SAM value increase of 0.0009, ERGAS increase of 0.0166, SCC value decrease of 0.0001, and $Q2^n$ value decrease of 0.0006. This indicates the effectiveness of DUDB-DBFM in fusing information from multiple domains and ensuring the integrity of spectral and spatial information. Also after calculation, the number of learnable variant parameter counts for ablation experiments is less than that of DUDB, which explains their performance degradation. In conclusion, the second set of ablation experiments validates the practicality of the proposed DUDB module.

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